numbers between iterations to give a small computational saving. We can see that only one vector of random numbers is generated per iteration, with the new vector of random numbers being used in the following iteration. This gives some correlation between steps that complicates our analysis a little.

If we wish to find the perfect sampling bias in the case of the one-dimensional harmonic oscillator, we can apply this technique to find the invariant averages under this scheme. We have

$$\Psi^{\mathrm{BBK}^*} = \begin{bmatrix} 1-\frac{h^2\Omega^2}{2m} & \frac{h}{m}\left(1-\frac{\gamma h}{2}\right) \\ \frac{h\Omega^2}{2m}\left(\frac{h^2\Omega^2-4m}{2+\gamma h}\right) & \frac{(2-\gamma h)\left(2m-h^2\Omega^2\right)}{2m(2+\gamma h)} \end{bmatrix},$$

$$\mu_n^{\mathrm{BBK}^*} = \begin{bmatrix} \sqrt{\frac{k_\mathrm{B}\mathrm{T}\gamma h^3}{2m}}\mathrm{R}_n \\ \frac{\sqrt{2k_\mathrm{B}\mathrm{T}\gamma hm}}{2+\gamma h}\left(\left(1-\frac{h^2\Omega^2}{2m}\right)\mathrm{R}_n+\mathrm{R}_{n+1}\right) \end{bmatrix} = \begin{bmatrix} \kappa_1\mathrm{R}_n \\ \kappa_2\mathrm{R}_n+\kappa_3\mathrm{R}_{n+1} \end{bmatrix},$$

using the notation in (7.7). Taking expectations of the resulting equations, we can compute the values of the constants in (7.8)–(7.10) and solve to find the long-time averages. Care must be taken however, as unlike the schemes using the ABO splitting we must include some correlation between steps. Denoting the entries of Ψ^{BBK^*} by ψ_{ij} we have

$$\begin{aligned}\mathbb{E}(\mu_{n,2}p_n) &= \mathbb{E}\left(\left(\kappa_2\mathrm{R}_n+\kappa_3\mathrm{R}_{n+1}\right)\left(\psi_{21}q_{n-1}+\psi_{22}p_{n-1}+\kappa_2\mathrm{R}_{n-1}+\kappa_3\mathrm{R}_n\right)\right), \\ &= \kappa_2\kappa_3.\end{aligned}$$

This additional quantity will slightly change the expectation of (7.9) compared to the standard BBK method, and hence the overall averages of the scheme. Additionally it is easily shown that

$$\mathbb{E}(\mu_{n,1}q_n) = \mathbb{E}(\mu_{n,2}q_n) = 0, \quad \mathbb{E}(\mu_{n,1}p_n) = \kappa_1\kappa_3,$$

with

$$\mathbb{E}(\mu_{n,1}^2) = \kappa_1^2, \quad \mathbb{E}(\mu_{n,1}\mu_{n,2}) = \kappa_1\kappa_2, \quad \mathbb{E}(\mu_{n,2}^2) = \kappa_2^2+\kappa_3^2.$$

Thus taking expectations of (7.8)–(7.10) and substituting in the constants gives

$$\begin{aligned}\langle q^2\rangle_h^{\mathrm{BBK}^*} &= \psi_{11}^2\langle q^2\rangle_h^{\mathrm{BBK}^*}+\psi_{12}^2\langle p^2\rangle_h^{\mathrm{BBK}^*}+2\psi_{11}\psi_{12}\langle qp\rangle_h^{\mathrm{BBK}^*}+2\psi_{12}\kappa_1 2\psi_{12}\kappa_1\kappa_3 \\ &\quad +\kappa_1^2, \\ \langle p^2\rangle_h^{\mathrm{BBK}^*} &= \psi_{21}^2\langle q^2\rangle_h^{\mathrm{BBK}^*}+\psi_{22}^2\langle p^2\rangle_h^{\mathrm{BBK}^*}+2\psi_{21}\psi_{22}\langle qp\rangle_h^{\mathrm{BBK}^*}+2\psi_{22}\kappa_2 2\psi_{22}\kappa_2\kappa_3 \\ &\quad +\kappa_2^2+\kappa_3^2,\end{aligned}$$